Distributive Property
In equation form, the distributive property is written:

EQUATION 7-8
The distributive property of convolution describes how parallel systems are analyzed.

$$a[n] * b[n] + a[n] * c[n] = a[n] * (b[n] + c[n])$$

The distributive property describes the operation of **parallel systems with added outputs**. As shown in Fig. 7-10, two or more systems can share the same input, $x[n]$, and have their outputs added to produce $y[n]$. The distributive property allows this combination of systems to be replaced with a single system, having an impulse response equal to the *sum* of the impulse responses of the original systems.

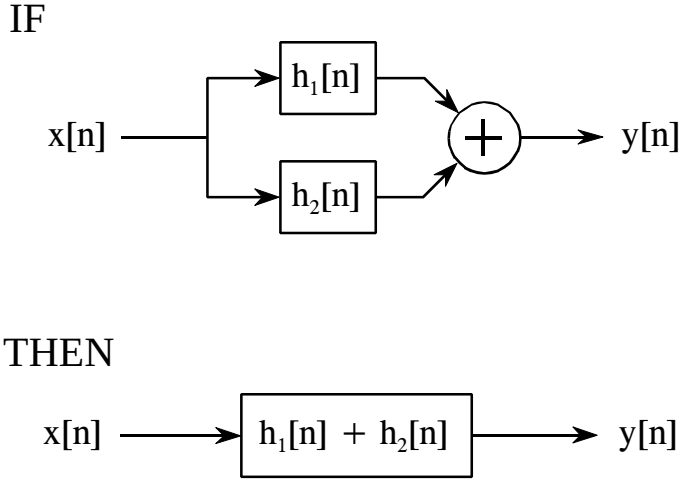

FIGURE 7-10
The distributive property in system theory. The distributive property shows that parallel systems with added outputs can be replaced with a single system. The impulse response of the replacement system is equal to the sum of the impulse responses of all the original systems.

Transference between the Input and Output
Rather than being a formal mathematical property, this is a way of thinking about a common situation in signal processing. As illustrated in Fig. 7-11, imagine a linear system receiving an input signal, $x[n]$, and generating an output signal, $y[n]$. Now suppose that the input signal is changed in some linear way, resulting in a new input signal, which we will call $x'[n]$. This results in a new output signal, $y'[n]$. The question is, how does the change in